

Market Microstructure and Electronic Trading Finance 480

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Market Structures

Recall last lecture we discussed trades and traders:

- Post-trade order details: settlement and delivery;
- Connection between delivery and storage;
- Data issues with using trades for research;
- Different types of traders; and,
- The differing effects these traders have on markets.

Readings on Today's Topic: Market Structures

- Chapter 3 of Johnson.
- Blackboard readings.

If you are reading Harris, a few more comments:

- p. 100: I always knew QQQ as the “Q’s”, not the “cube²”.
- p. 103: QuantEx was out-of-date in 2003³.

²Journalists and non-traders seem to be the source of this term.

³ITG's idea won; they lost.

Introduction

- Today we'll discuss different continuous market structures.
- Classically: order- versus quote-driven markets.
- Related: difference between specialist and market maker.
- I'll mention crossings and dark pools.
- After the break, we will focus on decentralized markets.
- I'll mention benefits and challenges of these markets.
- I'll also mention Reg NMS: what it is and why you should care.
- Finally, think about the CLOB in these markets.

- A term we'll be using a lot: *latency*.
- Latency: Time delay in sending a message (quote/order/fill).
- Often: delay from send to fill receipt for marketable order.
- Low latency is nice:
 - Freshest data + fast fills = best prices.
- Low latency increases robustness, reduces opportunity costs:
 - Quickly know if filled and can try elsewhere.
- Don't need to hold positions briefly for latency to matter.

Throughput

- Another term to consider: *throughput*.
- Plainly: how many messages can flow through a system.
- High throughput is good; allows very frequent trading.
- However, throughput has a dark side:
 - All venues have some limit to their throughput;
 - However, messages are usually not tolled.
 - Could send more messages than can be handled — for free.
 - Some venues already experience this (“quote stuffing”).
- Hasbrouck (2010?) suggests possible denial-of-service attacks.

Continuous Market Structures

Continuous Market Structures

- Continuous markets historically broken into three types:
 - Order-driven markets;
 - Quote-driven markets; and,
 - Brokered markets.
- The difference is on who creates liquidity:
 - Traders and investors (order-driven);
 - Market makers (quote-driven); and,
 - Brokers (less liquid; often OTC).
- In a way, this distinction is artificial.
 - First two types may not be that different.
 - Many markets are *hybrids* of these types.

- Many stock exchanges started as order-driven markets.
- One problem: Orders might not always match:
 - Market would then function only sporadically.
 - Lower volume would dissuade investors from trading there.
- Solution? “Hire” someone to watch market:
 - Ensure it is “fair and orderly.”
 - Constantly quote a bid and ask.
 - Also provide liquidity as needed.
- This person is called a *specialist*.

Specialists: How

- I just pulled a fast one on you.
- Specialist watches the market. How?
 - Historically, only specialist could see the book.
 - Specialists also had/have a latency advantage.
 - These are huge informational advantages
- Thus specialists tend to fight dissemination of the book.
- When specialists trade, they change their inventory⁴.
- Thus specialist takes on market and adverse selection risks.

⁴Example on board.

Specialists: Why

- Specialists have position⁵. Why take such risk?
- Specialists also step in when most orders are same-side.
 - This situation is like a one-sided auction.
 - The specialist can earn the winner's curse.
- Specialists may even earn twice that. For example:
 - Specialist is long 1000 shares bought at \$10.
 - Price spikes from \$10 to \$12; some buys still unfilled.
 - Specialist sells 2000 shares⁶ at \$14.
 - If buying pressure is satisfied, price eases back to \$10.
 - Specialist makes \$4/share on 1000+2000 shares (\$12k).

⁵*i.e.* Inventory.

⁶Sell 1000 shares long and 1000 shares short.

Specialists: Why Now?

- Possible problems for specialists in modern markets:
 - Order books may be disseminated widely.
 - Many traders have low-latency data and execution.
 - ECNs: counter-argument to needing specialist.
- But then who provides liquidity?
 - Other traders (*i.e.* the market).
- NYSE has baroque systems to keep specialist fast enough:
 - e.g. Direct+ (Auto-X on quote)

Specialists: For A Few Dollars More



Clorox (CLX) on 20080915 close and 20080916 open.

- Suppose there is heavy selling on the close.
- 80,000 shares will pair off; 9,200 MKT sells unfilled.
- Specialist steps in to buy 9,200 shares.
- \$63.28 at 15:58; \$63.05 at close; \$63.48 at 9:42 (next day).
- Profit: $9,200 \times \$0.43 = \$3,956$.
- So now you know how to do this. Except...
- NYSE doesn't publish imbalances in real-time.

- Contrast the specialist with the market maker (MM):
 - Specialist chooses when to step in; has advantages.
 - MM has no advantage on discerning when to step in.
- Quote-driven markets are thus different.
 - No monopolies on all information (maybe on their own LOB).
 - But, MMs may be given preferential treatment.
- Thus MMs regularly face a tough optimization problem.

Market Makers: Always Be Optimizing⁷

- Due to quoting, MMs usually have inventory (net long/short).
- The good: Make the spread.
- The bad: Cannot stop quoting; only change prices and sizes.
- The ugly: Exposed to adverse selection, market moves.
- This leads to a continuous optimization:

Max (Spread Revenue) - (Market Move Losses)
Subject to (Position/Risk Limits)

⁷Like “Always be closing,” with less getting fired.

Designated Primary Market Makers

- CBOE uses a hybrid of a specialist and MM.
- *Designated primary market makers* (DPMs).
- DPMs are like specialists:
 - See whole order book;
 - Charged with maintaining fair and orderly market.
- But, there are also other MMs.
- In fact, DPMs were added after multiple MMs already existed.
- Likely explanation for unavailability of CBOE order book.

Market Makers: Membership Has Its Privileges

- Anyone can act as MM; official MMs have certain freedoms:
 - Sell short without price tests.
 - Need not worry strictly about locates.
 - Lowest latency view of/access to their own book.
- Official MMs also have certain responsibilities:
 - Must always supply two-sided quote.
 - Must honor quotes. (No “backing away.”)
 - Fill order over quoted size or move quote. (“Trade or fade”)
 - Violation: temporarily suspended from making that security.
 - Fines for repeated violations.
- Thus quotes tend to be kept current and updated often.

- So it would seem quotes are ideal for inference.
- Except quotes have a few problems:
 - May be wide;
 - Reflect (unseen) inventory risk;
 - Only bound where a security *could* trade;
 - May be far from where securities would trade.
- Lining up trades and quotes in non-trivial.
- Even lining up quotes from different venues is tough.

Electronic Communications Networks (ECNs)

- First electronic communications network (ECN): 1970.
 - Instinet created by institutional traders.
 - Way to trade large volume without (feared) market makers.
 - Also eliminated paying MM/broker full spread⁸.
- An ECN is a limit order book that:
 - Displays orders to the outside world;
 - Matches orders in some priority;
 - Sends fills to involved traders; and,
 - Sends execution prints to the tape.
- ECNs mostly make money from usage fees.

⁸Even in 1990s, brokers kept spreads artificially high.

ECNs: Advantages

Initially apparent:

- Need not pay spread; see price you will get (low uncertainty).

Apparent later:

- Required tasks (for ECN) are computationally light.
- ECN software: generations ahead of other venues.
- Thus ECNs can handle large order volumes.
- ECNs also became early proponents of low-latency trading.
 - In 2000: NYSE fill $\approx 3s$; Island fill $\approx 300ms$.
 - In 2003: Island fill $\approx 20-30ms$.
 - In 2012: BATS fill $\approx 500 \mu s$.⁹

Crossings and Dark Pools

- Harris, typical investments courses, HBS case:
 - All mention ITG's POSIT system.
 - POSIT runs scheduled crosses of orders a few times/day.
- Other venues also offer crossing.
- Many see crossing as beneficial:
 - Can send large orders without showing size; and,
 - Can be used without needing major infrastructure.
- *Dark pools* are venues with undisplayed liquidity.
 - Do not participate in NBBO, CLOB.
 - Thus POSIT is a dark pool (since 1987).

Problems with Crossings and Dark Pools

I don't care to belong to a club that accepts people like me as members. — Groucho Marx

- Problem with crossings and dark pools: adverse selection.
 - Classic game theory example: Paradox of Trade.
- Idea: Bad bets get filled; good bets don't.
- Problem is not unique to dark pools.
- Why it is worse for dark pools:
 - You remove your trades from visible market.
 - Lose order protection, ease of hunting down a fill.

Decentralized Markets

- Decentralization requires changing our perspective:
 - Often complicates classifying a market;
 - Has some clear and powerful advantages; but,
 - Has some subtle or nasty disadvantages.
- Recent regulation tried to fix problems yet retain benefits.
- Decentralization also complicates research:
 - Some research approaches become intractable.
 - Long-held microstructure assumptions may be broken.

MMs and ECNs

- Basic difference: quote- versus order-driven.
- However, in decentralized markets we may have both.
 - MMs may quote inside an ECN.
 - ECNs may route orders to MMs.
- Another difference:
 - MMs may trade for own account; have inventory.
 - ECNs cannot trade for selves; always flat.
- ECNs cannot take risk to offer negotiated trading.
 - MMs may fill at close — even without an auction.
- MMs need to trust their counterparties.
 - May auto-execute¹⁰ more than displayed size for trusted clients.

¹⁰Often referred to as “auto-X.”

Decentralization Advantages

- Main advantage of decentralization: competition.
- Competition has brought many key changes:
 - Forced exchanges to reduce latency;
 - Made certain features/data expected (e.g. LOB);
 - Lowered trading costs¹¹; and,
 - punished cumbersome order/trading rules¹².
- Also has made market more robust to exchange problems¹³.

¹¹Lower spreads; decimalization.

¹²AMEX; early Instinet.

¹³Nasdaq crash on 2002 Russell rebalance.

Decentralization Disadvantages

- Decentralization fractures liquidity.
 - Liquidity tends to attract more liquidity.
- Increases likelihood of being *traded through*.
 - Execution at another venue would have filled your order.
- Can make market more expensive to trade:
 - Need to monitor N venues;
 - Maintain connections to each venue; and,
 - Route orders to appropriate venue.
- Unsophisticated customers likely ignorant of these issues.

Infrastructure Challenges

- Already hinted at some challenges of decentralization.
- Ideal: a unified view of the market (“market mosaic”).
- Problems with creating a mosaic:
 - Clock differences between venues¹⁴;
 - Latency differences (some “tiles” may be out-of-date);
 - Functionality differences between venues; and,
 - Reliability differences.
- Building most current mosaic possible can be painful:
 - Direct feeds from/connections to each venue;
 - Constantly deciding how to split, route, re-route orders.

¹⁴Critical for trading up to the close.

- In decentralized market, easy questions are hard to answer:
 - Did you get best execution possible?
 - What was NBBO at time t ?
 - What NBBO did someone see at time t ?
 - Where do I send this order?
 - Should I send venue A more flow next hour? next minute?
- Slow traders may be increasingly disadvantaged.
 - May not even know how to measure their disadvantage.

Reg NMS: Introduction

- To address issues, SEC adopted Reg NMS in June 2005.
- Idea: Create a national market system.
- Effective date: Various rules phased in 2006–late 2007.
- Composed of four main rules:
 - Order Protection Rule
 - Access Rule
 - Sub-penny Pricing Rule
 - Market Data Rules and Plans
- Also included prior rules about market center reporting:
 - Publish standard execution quality and routing reports.

- Order Protection Rule

- All venues must display NBBO;
- Cannot trade though NBBO; and,
- Discretionary orders only protected at displayed price.

- Access Rule

- Requires private intermarket links for quote access;
- SROs must police disseminating quotes which lock/cross;
- Limits fees for taking liquidity; and,
- Venues must display quotes if $> 5\%$ of ADV.

- Sub-penny Pricing Rule
 - No orders priced in increments less than \$0.01.
 - Idea: Increase cost of “stepping ahead” of orders.
- Market Data Rules and Plans
 - Allocates SRO MD fees based on quote/execution quality.
- Execution and Routing Reporting Rules (Rules 605, 606)
 - Mandates standardized statistical reports be made public.
 - Monthly: Order executions (speed and price) across spread.
 - Quarterly: Order routing and payment-for-flow agreements.

¹⁵E.B.

- Reg NMS stopped short of requiring a CLOB.
- Requirements to achieve synchrony would have been onerous.
- For now, anti-competitive aspects of CLOB made clear.
- But requirements to route away¹⁶ have a cost (esp. speed).
- Some venues find this cost intolerable.
- One way out: go “dark.”

¹⁶Send orders elsewhere for a better price.

- Island protested intermarket linkage in 2002.
 - Stopped disseminating book for three ETFs.
 - Most active ETFs; Island was dominant venue.
- Effects were sudden and strong:
 - Island's market share fell (high 50%s ↓ mid 30%s);
 - Prices adjusted more slowly (quote changes became AR);
 - Price discovery shifted to futures; and,
 - Trading costs increased overall.
- Island later re-illuminated; execution quality then increased.

¹⁷Hendershott and Jones (2005).

SuperMontage: CLOB Lite?

- In 2001 Nasdaq released plans for SuperMontage¹⁸
 - Incorporated all participants LOBs;
 - Consolidated quote display and execution systems;
 - Allowed for routing or not;
 - Attempted to address routing speed (via NOW orders); and,
 - Created anonymous order source mnemonic (SIZE).
- Some venues complained the idea was pointless:
 - Not low-latency; why participate?
- Some low-latency venues (e.g. BATS) left SuperMontage.

¹⁸TotalView is the latest version of SuperMontage.

De Facto Privatization of CLOB

- Need for low latency has major effects on infrastructure.
- For best mosaic, Reg NMS and SuperMontage are insufficient.
- Many market participants have done for themselves:
 - Direct feeds from venues;
 - Direct market access (DMA) to venues;
 - Co-location of participant and venue servers; and,
 - Smart order routing.
- Market participants have built own CLOBs of varying quality.

Listing Versus Traded Market

And now for something completely different.

- We've broken an assumption here; and, nobody's admitting it.
- Researchers talk about "NYSE" or "Nasdaq" stocks.
- Those designations refer to where the stock is listed.
- Usually, securities trade primarily on their listing market.
- In a decentralized market, this may not hold.
- Which microstructure effects are:
 - Due to listing market?
 - Due to trading market?
- Little thought has been given to this issue.
- Similar to creating futures series from near+next and roll?

The Road Ahead

We've covered the basics and hinted at the complex world of market structures.

Now we move on to more quantitative fare.

Hopefully, this will help you gain trading intuition, because... then we will start putting the pieces together.

- Metrics/Models: Prices; Sizes; Limits; Time; Liquidity
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

Readings For Next Lecture

Readings, in order of importance:

- Chapters 3, 5, and 6 of Hasbrouck.
- Chapter 3 of Foucault, Pagano, and Röell.
- Blackboard: Readings